Does Graph Smoothing Reintroduce Bias into Fair Node Embeddings? An Exact-Inference Audit on Pokec-z

ICT 5143: Algorithms Fairness and Graphical Models Laboratory
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Abstract

We ask whether exact pairwise Markov smoothing undoes the fairness gains of FairWalk embeddings in Pokec-z working-field prediction. On a fixed 2,000-node subgraph (5 seeds, 1,000 graph-block bootstrap replicates) uniform smoothing changed the gender equal-opportunity gap of the FairWalk pipeline by 0.002 [-0.038, 0.022] (Holm $p = 1.000$), and our homophily-constrained edge attenuation did not differ from a global scalar (0.000 [-0.037, 0.023]). The core graph is gender-heterophilous, so the mechanism is absent. In simulation, smoothing widened group gaps only when demographic homophily and a base-rate gap were both present, and a global scalar matched HCA there too. Findings are subgraph-specific.

1 Question, context and contribution

Stakeholders are social-platform users, graph-learning researchers and fairness auditors. $Y{=}1$ is the FairGNN benchmark's positive working-field class (raw code ≥ 1; meaning of codes undocumented); $D{=}1$ means the pipeline predicts that class. This is an offline representation audit, not a recruitment system. **Research question:** when does probabilistic graph smoothing undo the fairness gains of debiased embeddings, and can constrained edge potentials reduce this effect? We expected uniform smoothing to widen the gender gap in true-positive rates (ΔEOpp) under demographic homophily and a base-rate gap.

Contribution: (i) a controlled, exact-inference decomposition of fairness change between the embedding stage and a downstream conditional Markov network (CMN), on real and simulated graphs; (ii) homophily-constrained attenuation (HCA) $\lambda_{ij} = \lambda(1-\eta\,\mathbf{1}[A_i{=}A_j])$, refuted if it does not beat a global scalar with the same tuning budget (H4).

Closest work: FairWalk [1] (reproduced, restricted re-implementation) mitigates at the embedding stage; Correct&Smooth [3] smooths a graph-agnostic predictor without fairness analysis; FairDrop [4] attenuates homophilous edges during representation learning, not in downstream potentials. Twelve verified papers: `docs/related_work.md`.

2 Data, graphical model and fairness method

Data and estimand: Pokec-z from the FairGNN repository [2] (SHA-256 recorded; accessed 2026-11-05). Target reproduced from FairGNN code: code -1 unlabelled, $0 \to Y{=}0$, $1\text{–}4 \to Y{=}1$. Protected attribute: recorded gender (0/1; SNAP documents 1 as "man"); region is a secondary context dimension. Core: labelled-node largest component, BFS ball of 2,000 nodes (seeded, label-value blind): 5,351 edges, $P(Y{=}1)$ 0.40 (gender 0) vs 0.56 (gender 1). Edge homophily: gender 0.449 (random mixing 0.500), region 0.963 (0.514), label 0.586 (0.501). Transductive 50/25/25 label splits stratified by $Y\times$gender; the split unit for uncertainty is a Louvain graph block.

Graph and inference: unary ψ_i from an L2 logistic regression on Node2Vec or FairWalk embeddings H_i (fitted on training labels); training labels enter as indicator evidence; validation/test labels are latent. On a label-free maximum-Jaccard spanning forest E_F (1999 edges, 37.4% kept):

$$P(\mathbf{Y} \mid \mathbf{H}, \mathbf{X}, E_F) = \frac{1}{Z} \prod_i \psi_i(Y_i; H_i) \prod_{(i,j)\in E_F} e^{\lambda_{ij}\mathbf{1}[Y_i=Y_j]}, \qquad q_i = P(Y_i{=}1 \mid \mathbf{y}_{\text{train}}),\ D_i = \mathbf{1}[q_i \geq \tau].$$

q_i is computed by exact log-space sum-product (upward/downward passes per tree). The Markov blanket of Y_i is its forest neighbours; each positive neighbour adds λ_{ij} to the log-odds, which is how

homophilous neighbourhoods can import group base rates.

Method and fairness: λ is fitted on five training folds (held-out fold latent, cross-fitted unaries); η (HCA) and the global scale c (ablation) are chosen on validation by the same rule: minimise ΔEOpp subject to macro-F1 $\geq$ uniform -0.02. Primary criterion: gender $\Delta\text{EOpp} = \max_g t_g - \min_g t_g$, because base rates of the recorded label differ by gender and the concern is unequal errors among true positives introduced by smoothing; ΔDP, ΔEOdds and intersectional gaps are diagnostics. Primary utility: macro-F1; acceptable loss 0.02. HCA needs A at inference time (FairWalk only at representation time; group thresholds at decision time).

3 Evaluation protocol

Baselines: feature-only LR, SGC (2-hop simple GNN), Node2Vec+LR, FairWalk+LR, per-gender thresholds on Node2Vec (decision-only), and the same CMN with uniform smoothing. Ablations: no pairwise factors, global scalar, Node2Vec instead of FairWalk. All methods share graph, splits, forest, threshold rule (validation-maximal macro-F1) and an 11-point tuning grid; matched-utility operating points use a common validation floor. Five seeds (11–55) refit splits and embeddings. Fixed-model uncertainty: 1,000 replicates resampling 15 Louvain blocks (inter-block edge share 0.264) with identical draws for all methods; paired effects averaged over seeds; percentile 95% intervals. Confirmatory family H1–H4 (ΔEOpp, Holm), declared before test evaluation; everything else exploratory. Synthetic variability (10 realisations/cell) is reported separately. Exact inference: enumeration, forest sum-product, variable elimination and clique tree agree to 1.6e-15 on 200 random fixtures (tolerance 10^{-8}). Mandatory experiments 1–5, ablations and five robustness tests: `docs/requirements_matrix.md`.

4 Results and interpretation

Test, mean of 5 seeds [95% block CI]	macro-F1	ΔEOpp gender	ΔDP gender	Brier
Node2Vec+LR (unmitigated)	0.667	0.101 [0.057, 0.157]	0.034	0.214
FairWalk+LR (embedding-only)	0.657	0.087 [0.057, 0.157]	0.032	0.215
FairWalk+uniform CMN	0.669	0.089 [0.045, 0.157]	0.032	0.211
FairWalk+global scalar (abl. B)	0.660	0.092 [0.053, 0.160]	0.035	0.214
FairWalk+HCA (proposed)	0.660	0.092 [0.041, 0.161]	0.038	0.212
Node2Vec+group thresholds	0.667	0.013 [0.020, 0.148]	0.098	0.214

Table note: About 500 test nodes per seed; every gender and gender×region cell has $\geq$33 positives and negatives in every seed (none below the 30-observation flag). Intervals for absolute gaps are percentile intervals of |gap| and are pushed away from zero, so a near-zero estimate can fall below its interval (group thresholds). Default (validation-maximal macro-F1) thresholds. Paired effects (mean over seeds [95% block CI], Holm p): H1 FairWalk uniform$-$LR 0.002 [-0.038, 0.022], p=1.000; H2 Node2Vec 0.001 [-0.028, 0.039], p=1.000; H3 HCA$-$uniform 0.004 [-0.020, 0.026], p=1.000; H4 HCA$-$global 0.000 [-0.037, 0.023], p=1.000.

Main finding: on Pokec-z, smoothing did not undo or create a gender fairness difference: all four confirmatory effects are near zero with intervals spanning roughly $\pm$0.03. Smoothing improved utility and calibration slightly (Node2Vec: macro-F1 +0.011 [0.003, 0.022], Brier -0.004 [-0.007, -0.001]; exploratory). FairWalk also had little to remove: gender was barely predictable from either embedding (AUROC 0.511 vs 0.501), whereas region was (0.991), so lower attribute predictability did not translate into a measurably fairer decision (ΔEOpp difference -0.015 [-0.039, 0.032]).

Mechanism and failure: the ablation without pairwise factors reproduced the unary predictions exactly, so all changes are due to ψ_{ij}. The fitted coupling was weak ($\lambda = 0.25$ in every seed): the forest is somewhat label-homophilous (0.629) but keeps only 37.4% of edges, (mean forest degree 2.0), so the smoothing signal itself was small. In simulation with a fair-by-construction unary, uniform smoothing raised ΔDP by 0.066 ($\pm$0.038 SD over realisations) at demographic homophily 0.9 but -0.009 at 0.5 (no homophily), and only 0.014 at 0.9 without a base-rate gap. Across all 15 homophily×degree cells HCA$-$global on ΔEOpp ranged from -0.006 to 0.020: HCA never beat the scalar. Robustness (15

perturbed conditions × 5 seeds; dropout, corruption, rewiring, forests, dimensions, label budgets): the smoothing effect on ΔEOpp ranged from −0.027 to 0.019 for FairWalk (−0.021 to 0.037 for Node2Vec; seed means) with no consistent sign; the FairWalk utility gain held under edge perturbations but vanished with the three non-Jaccard forests. HCA's failure case: the validation-selected η was unstable across seeds (0.8, 0, 0.3, 0.5, 1), i.e. a 500-node validation set cannot reliably tune a fairness-targeted parameter whose test effect is within block-bootstrap noise. Decision-only thresholds cut ΔEOpp to 0.013 but raised ΔDP to 0.098. Full curves: `results/figures/`.

5 Limitations, ethics and conclusion

Real evidence comes from one BFS subgraph with gender heterophily; other subgraphs or Pokec-n may behave differently. Synthetic effects are properties of the simulator, not of Pokec. The forest keeps only 37.4% of edges; loopy BP on the full graph is only an approximate reference. Recorded gender is a binary profile field; region is predictable from embeddings, so releasing embeddings leaks region. HCA treats edges differently by group and is not justified by our evidence. The label has no defensible employment use. Conclusion: smoothing reintroduces bias only where neighbourhoods carry group base rates; on this Pokec-z subgraph they do not for gender, and HCA offers no advantage over simply weakening smoothing. Next experiment: repeat on region-homophilous subgraphs with region as the primary attribute.

Reproducibility and individual contributions

Reproduce: `scripts/run_experiments.py`, `run_synthetic.py`, `run_robustness.py`, `evaluate.py`, `make_results.py` with the frozen `configs/` (exact commands in `README.md`); environment `requirements-lock.txt` (Python 3.13, CPU).

Student 1: [concrete artifacts and the cross-check of Student 2's work].
Student 2: [concrete artifacts and the cross-check of Student 1's work].
Evidence: `docs/requirements_matrix.md` and `results/`.
AI assistance is disclosed in `docs/contributions.md`.

References

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